

File Packer & Unpacker with Encryption

Technology: Java Programming

Project Overview :-

This project is a **Java-based file utility tool** that provides functionality for **packing multiple files into a single archive** and **unpacking them back** with all metadata preserved.

To enhance security, the packed file is **encrypted**, and only authorized users can **decrypt & extract** the data.

The project also features a **Graphical User Interface (GUI)** for user-friendly interaction.

Key Features

• File Packing

- Combines multiple regular files into a single archive file.
- Stores metadata (file name, size, timestamp) along with file content.

• File Unpacking

- Extracts individual files from the packed archive.
- Restores all **original metadata** and file structure.

• Data Security

- Built-in **encryption and decryption** to protect packed files.

• Graphical User Interface (GUI)

- User-friendly GUI built in Java (Swing).
- Provides simple options for selecting files, encrypting, packing, and unpacking.

• Cross-platform

- Runs seamlessly on any system with a Java Runtime Environment (JRE).

Learning Outcomes

- Practical experience with Java I/O Streams and File Handling APIs.
- Implementation of metadata management during file operations.
- Strong understanding of encryption/decryption techniques in Java.
- Hands-on experience with GUI development (Swing).

- Insight into archiving and compression utilities (similar to ZIP/TAR).